

Requested exponents: mixed recovery, larger TP residuals

Step exponent 1 $\rightarrow$ 3/4; momentum exponent 0.6 $\rightarrow$ 1/2; smoothing stays 1/4. Scales and offsets were retained.

At 8 qubits and $\sigma = 0.05$, batch 65,536 falls from 99.0027% to 90.7749% fidelity at 148,000 steps.

Condition	Steps	Old F (%)	New F (%)	Δ F (pp)	Old loss	New loss	Old TP	New TP
n=4, β scale 10	10,000	1.2834	41.9235	+40.640	0.003675	0.002213	0.00245	0.02135
n=4, β scale 100	10,000	99.6561	99.6689	+0.013	0.001264	0.001266	0.002184	0.0205
n=4, β scale 1,000	10,000	99.3859	96.6945	-2.691	0.001277	0.00136	0.0122	0.02371
n=4, β scale 10,000	10,000	99.4292	96.2210	-3.208	0.001277	0.001372	0.09826	0.1709
n=5, β scale 10	10,000	0.1098	0.9729	+0.863	0.002062	0.002179	0.002092	0.01842
n=5, β scale 100	10,000	8.7965	9.1272	+0.331	0.001901	0.001891	0.002754	0.02083
n=5, β scale 1,000	10,000	72.6859	93.0621	+20.376	0.001388	0.001372	0.009505	0.02043
n=5, β scale 10,000	10,000	90.6799	89.2758	-1.404	0.001324	0.001405	0.07742	0.1421
n=8, B=8,192	200,000	90.8853	75.5377	-15.348	0.001135	0.001145	0.4835	0.8669
n=8, B=65,536	148,000	99.0027	90.7749	-8.228	0.001136	0.001135	0.199	0.3567
n=10, usual noise	100,000	0.004015	0.000011	-0.004	0.001162	0.001162	10.41	16.49
n=10, reduced noise*	57,000	99.0045	92.1621	-6.842	1.161e-06	1.207e-06	0.5105	0.9036

What changed. Fidelity improves in five of twelve comparisons, including n=5 at smoothing scale 1,000 (72.686% $\rightarrow$ 93.062%). The new n=4, scale-100 run retains 99.669%. Every new endpoint has a larger TP residual. Similar sampled losses can accompany very different recovery.

*Reduced-noise control: $\sigma = 0.0015625$ (32 times smaller); all other rows use $\sigma = 0.05$. F is normalized factor fidelity; TP is the unnormalized Frobenius residual. Δ F uses percentage points. These endpoints describe one dataset and seed, with no scale retuning.

Schedules and parameter choices

Update index k starts at 0. Iteration t is after t updates; the last applied update is $k=t-1$. Rank is 1 throughout.

Quantity	Old schedule	Requested schedule	Interpretation
Step γ_k	$\min(1, a/(k+b)^1)$	$\min(1, a/(k+b)^{3/4})$	Weight on the Frank-Wolfe atom
Smoothing β_k	$s/(k+1)^{1/4}$	$s/(k+1)^{1/4}$	TP penalty denominator; no cap
Momentum ρ_k	$\min(1, 2/(k+4)^{0.6})$	$\min(1, 2/(k+4)^{1/2})$	Weight on the new batch gradient

$d_k = (1-\rho_k)d_{k-1} + \rho_k \hat{g}_k$; the first gradient is assigned directly. The exact TP gradient is added after this averaging, divided by β_k . The factor update is $u_{k+1} = (1-\gamma_k)u_k + \gamma_k s_k$. The retained momentum weight is $1-\rho_k$.

Conditions	a / b	Smoothing scale s	Radius τ	Batch B	Fixed steps T
n=4, four scales	2 / 2	10; 100; 1,000; 10,000	10	32	10,000 each
n=5, four scales	2 / 2	10; 100; 1,000; 10,000	10	32	10,000 each
n=8, usual noise	10 / 10	10,000,000	20	8,192	200,000
n=8, usual noise	10 / 10	10,000,000	20	65,536	148,000
n=10, usual noise	10 / 10	1,000,000,000	40	65,536	100,000
n=10, reduced noise	10 / 10	1,000,000,000	40	65,536	57,000

The new late step sizes are much larger. For $n=8$ at $k=147,999$: γ changes from 6.75635×10^{-5} to 1.32521×10^{-3} (19.61 $\times$); ρ changes from 0.00158077 to 0.00519870 (3.29 $\times$). β stays 509,840.781. Thus the comparison changes effective update magnitudes as well as their decay rates.

The new step cap is active for $k=0, \dots, 11$ in $n=8/10$ and $k=0$ in $n=4/5$. The old step cap equals 1 only at $k=0$. The new ρ equals 1 at $k=0$; the first-gradient special case makes this inconsequential. Larger smoothing scale weakens TP enforcement throughout the schedule.

Fidelity, the objective, and the data budget

Fidelity is a separate recovery diagnostic; it is not f . With $\chi=uu^\dagger$ and synthetic truth $\chi^*=cc^\dagger$, $F(u,c)=|c^\dagger u|^2/(||c||^2||u||^2)$. This is the squared overlap of normalized complete factors, equivalently fidelity between their trace-normalized rank-one process matrices. Phase and nonzero scale do not affect it. High F alone does not certify trace preservation or normalization.

The $f+g$ formulation. $f(U)=(1/(2M)) \sum_s (A_s(UU^\dagger)-y_s)^2$ is measurement least-squares. $g(Z)$ is the indicator of $\{I\}$: zero at $Z=I$ and infinity otherwise, with $Z=T(UU^\dagger)$. The implemented smoothed objective is $f(U)+||T(UU^\dagger)-I||_F^2/(2\beta_k)$, over $||U||_{op} \leq \tau$.

Fidelity and TP use complete factors. Plotted loss and smoothed gap use the same fixed, independent sample of 512 rows (metric seed 12345). The noisy-truth expected loss is $\sigma^2/2$, not a measured floor or convergence certificate. Smoothed objectives/gaps at different smoothing scales are different penalized objectives.

Qubits / condition	Full row table $M=24^n$	Batch B	B / M (%)	B $\times$ T row draws	Draws / M (%)
4 (each scale)	331,776	32	0.009645062	320,000	96.45062
5 (each scale)	7,962,624	32	0.0004018776	320,000	4.018776
8, smaller batch	110,075,314,176	8,192	7.442177e-06	1,638,400,000	1.488435
8, larger batch	110,075,314,176	65,536	5.953742e-05	9,699,328,000	8.811538
10, usual noise	63,403,380,965,376	65,536	1.033636e-07	6,553,600,000	0.01033636
10, reduced noise	63,403,380,965,376	65,536	1.033636e-07	3,735,552,000	0.005891724

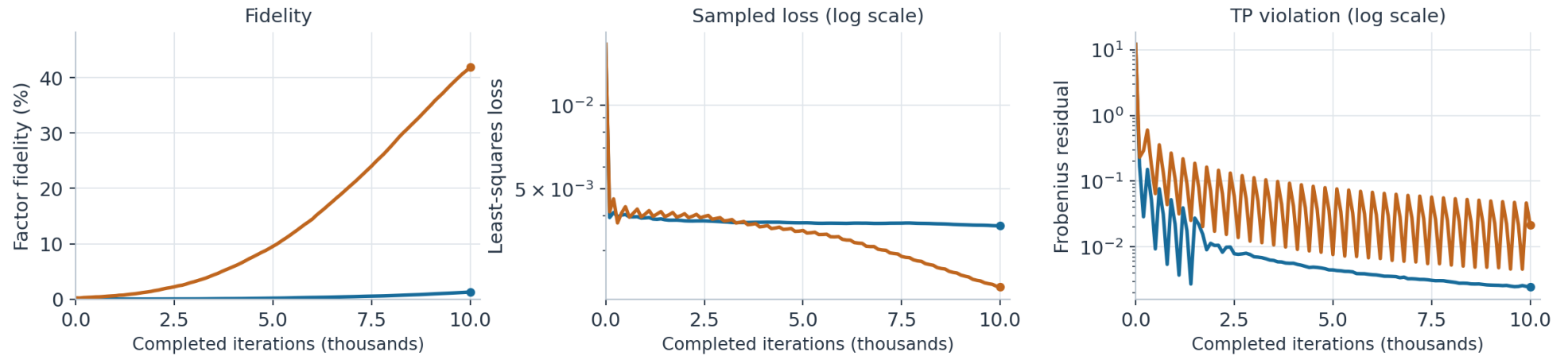
Rows are sampled uniformly with replacement. Draw counts include repeats and exclude metric/smoke-test work; they are **not unique-row coverage**. The $n=4/5$ compact datasets store all noisy observations. The $n=8/10$ table is virtual: sampled targets are generated from the truth and fixed row-dependent noise. Revisiting a row gives the same target. CPU preparation is prefetched for $n=8/10$.

Every condition uses Haar-unitary rank-one truth; channel, noise, initialization and sampling seeds are 0. All new runs start from the original U_0 and retain their own old dataset hashes. The on-demand noise recipe differs from stored-data generation, and fixed absolute σ does not imply fixed SNR across n . One dataset/seed per condition does not establish robustness or universal optimal settings.

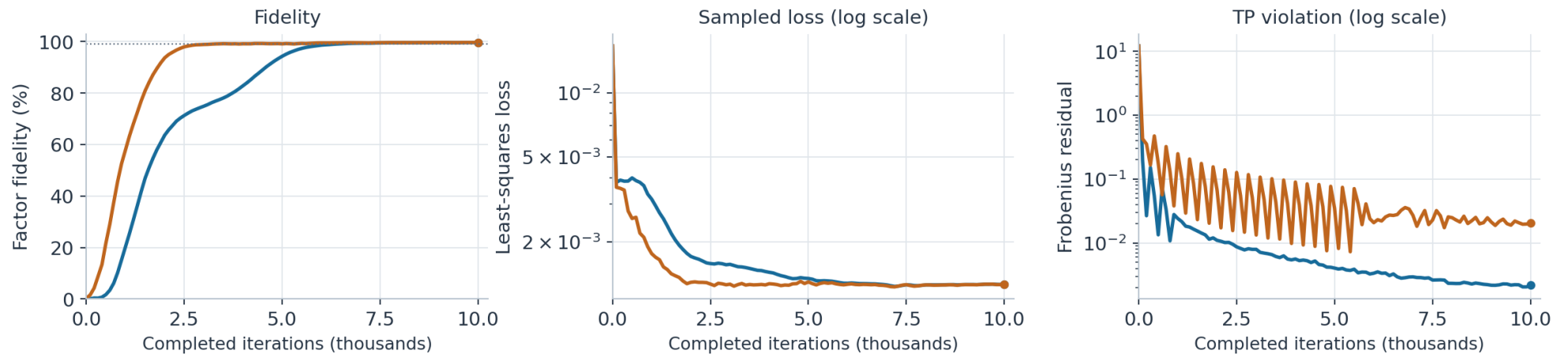
Four qubits: smoothing scales 10 and 100

— Old: step 1, smoothing 1/4, momentum 0.6 — New: step 3/4, smoothing 1/4, momentum 1/2

4 qubits | smoothing scale 10 | batch 32 | $\sigma = 0.05$



4 qubits | smoothing scale 100 | batch 32 | $\sigma = 0.05$



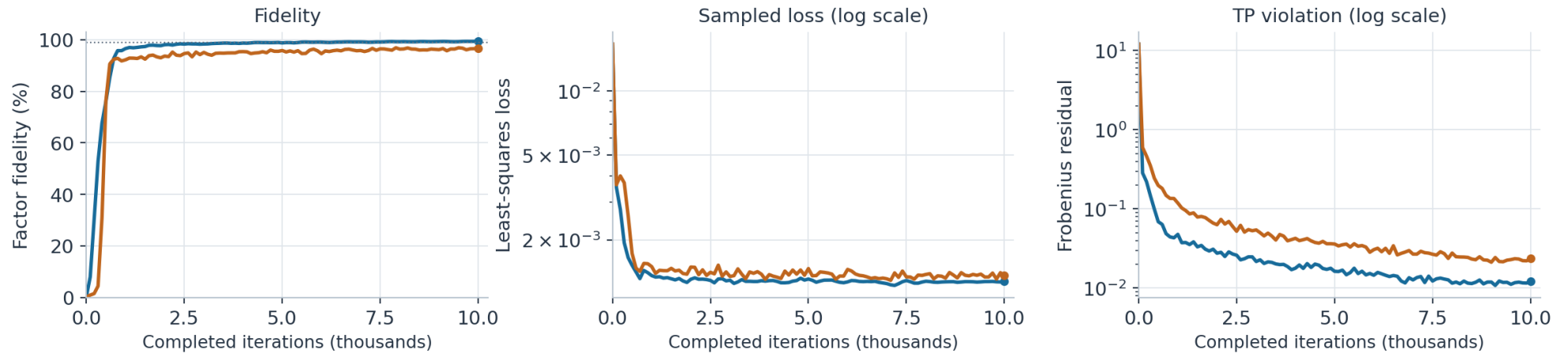
At scale 10, new fidelity improves substantially but remains far below 99%. At scale 100, both schedules exceed 99%; the new TP residual is about 9.4 times larger.

Fidelity and TP are complete-factor diagnostics. Loss uses 512 fixed sampled rows. Logarithmic loss/TP axes where indicated; dotted fidelity line marks 99% when in range.

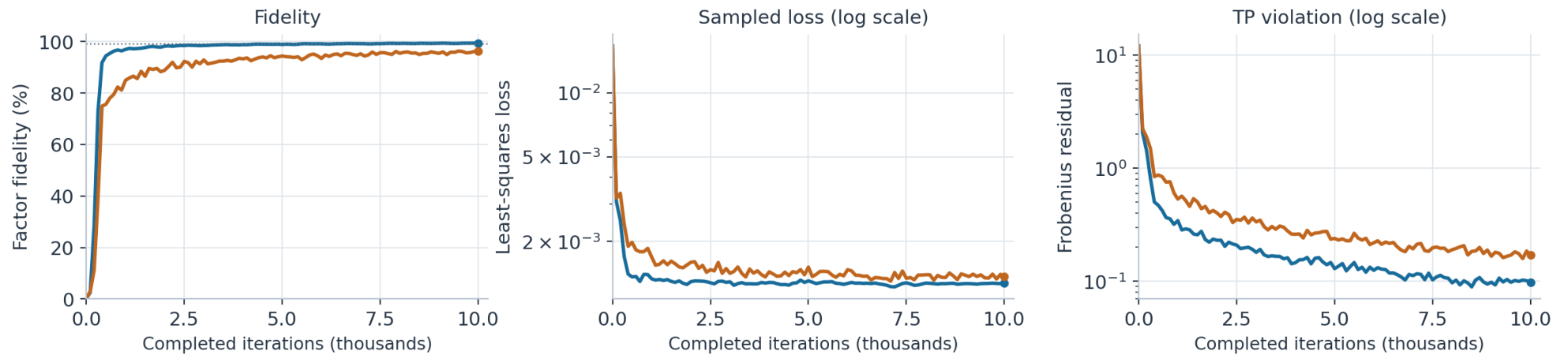
Four qubits: smoothing scales 1,000 and 10,000

— Old: step 1, smoothing 1/4, momentum 0.6 — New: step 3/4, smoothing 1/4, momentum 1/2

4 qubits | smoothing scale 1,000 | batch 32 | $\sigma = 0.05$



4 qubits | smoothing scale 10,000 | batch 32 | $\sigma = 0.05$



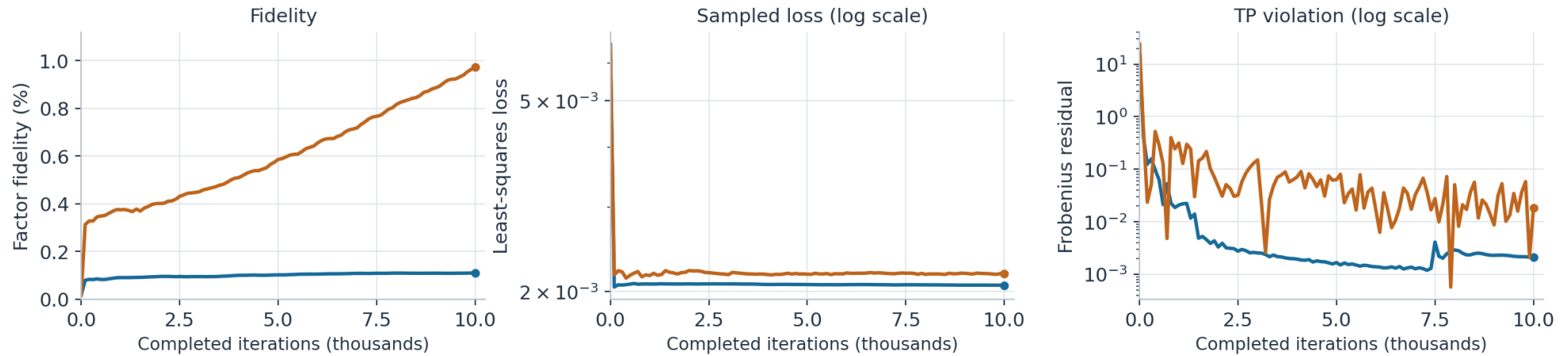
The old schedules exceed 99% at both scales. The requested schedules finish near 96%, with larger TP residuals. All curves show actual saved checkpoints, without fitting or smoothing.

Fidelity and TP are complete-factor diagnostics. Loss uses 512 fixed sampled rows. Logarithmic loss/TP axes where indicated; dotted fidelity line marks 99% when in range.

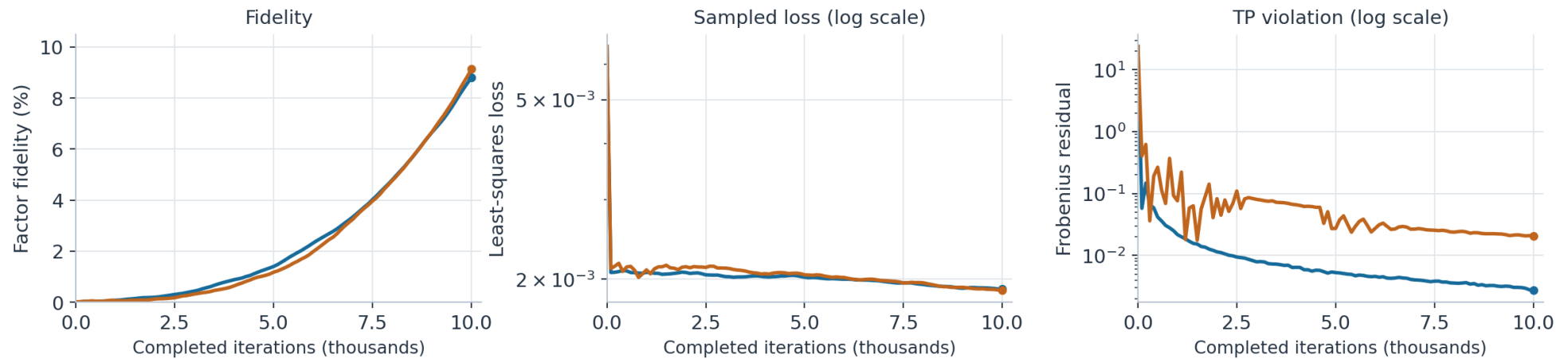
Five qubits: smoothing scales 10 and 100

— Old: step 1, smoothing 1/4, momentum 0.6 — New: step 3/4, smoothing 1/4, momentum 1/2

5 qubits | smoothing scale 10 | batch 32 | $\sigma = 0.05$



5 qubits | smoothing scale 100 | batch 32 | $\sigma = 0.05$



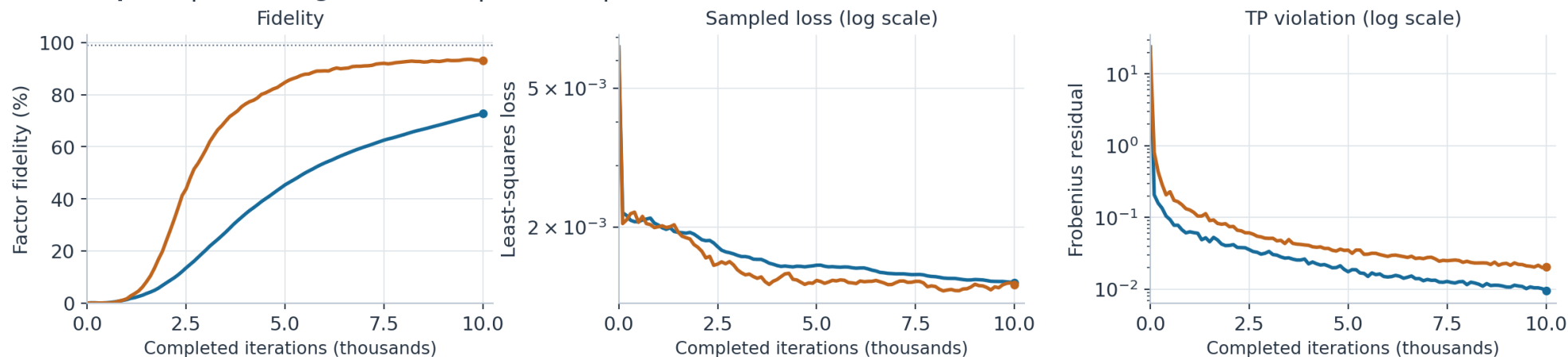
Fidelity axes are zoomed to show these low-recovery cases; neither approaches 99%. The requested schedules modestly improve final fidelity but worsen final TP.

Fidelity and TP are complete-factor diagnostics. Loss uses 512 fixed sampled rows. Logarithmic loss/TP axes where indicated; dotted fidelity line marks 99% when in range.

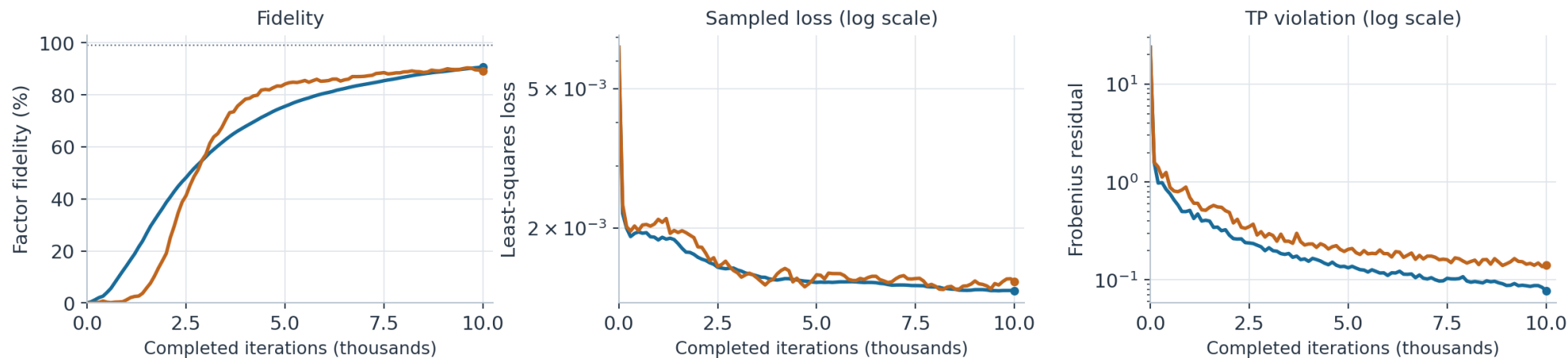
Five qubits: smoothing scales 1,000 and 10,000

— Old: step 1, smoothing 1/4, momentum 0.6 — New: step 3/4, smoothing 1/4, momentum 1/2

5 qubits | smoothing scale 1,000 | batch 32 | $\sigma = 0.05$



5 qubits | smoothing scale 10,000 | batch 32 | $\sigma = 0.05$



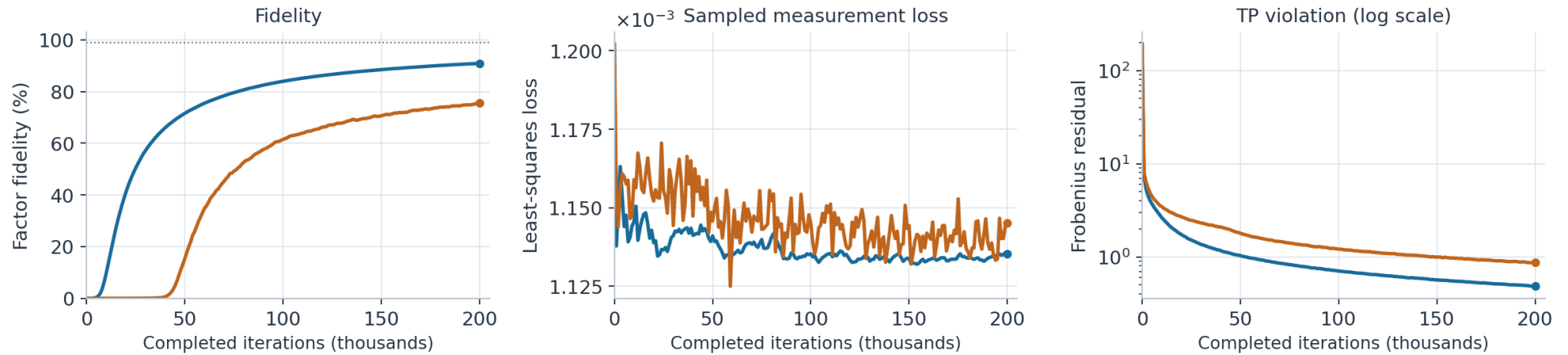
Scale 1,000 benefits most: final fidelity rises from 72.686% to 93.062%. Scale 10,000 falls from 90.680% to 89.276%. The new TP residual is larger in both.

Fidelity and TP are complete-factor diagnostics. Loss uses 512 fixed sampled rows. Logarithmic loss/TP axes where indicated; dotted fidelity line marks 99% when in range.

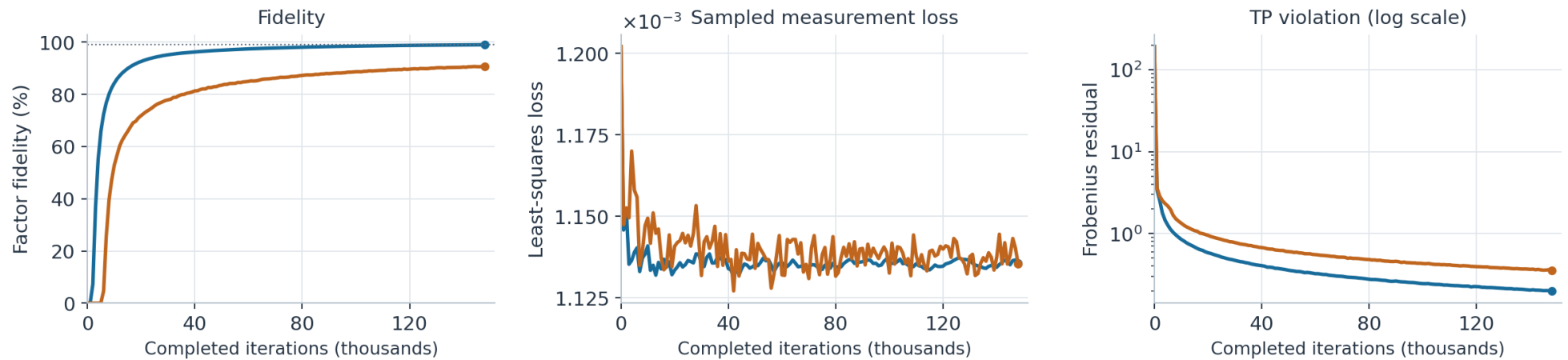
Eight qubits: usual noise, two batch sizes

— Old: step 1, smoothing 1/4, momentum 0.6 — New: step 3/4, smoothing 1/4, momentum 1/2

8 qubits | batch 8,192 | $\sigma = 0.05$ (usual noise)



8 qubits | batch 65,536 | $\sigma = 0.05$ (usual noise)

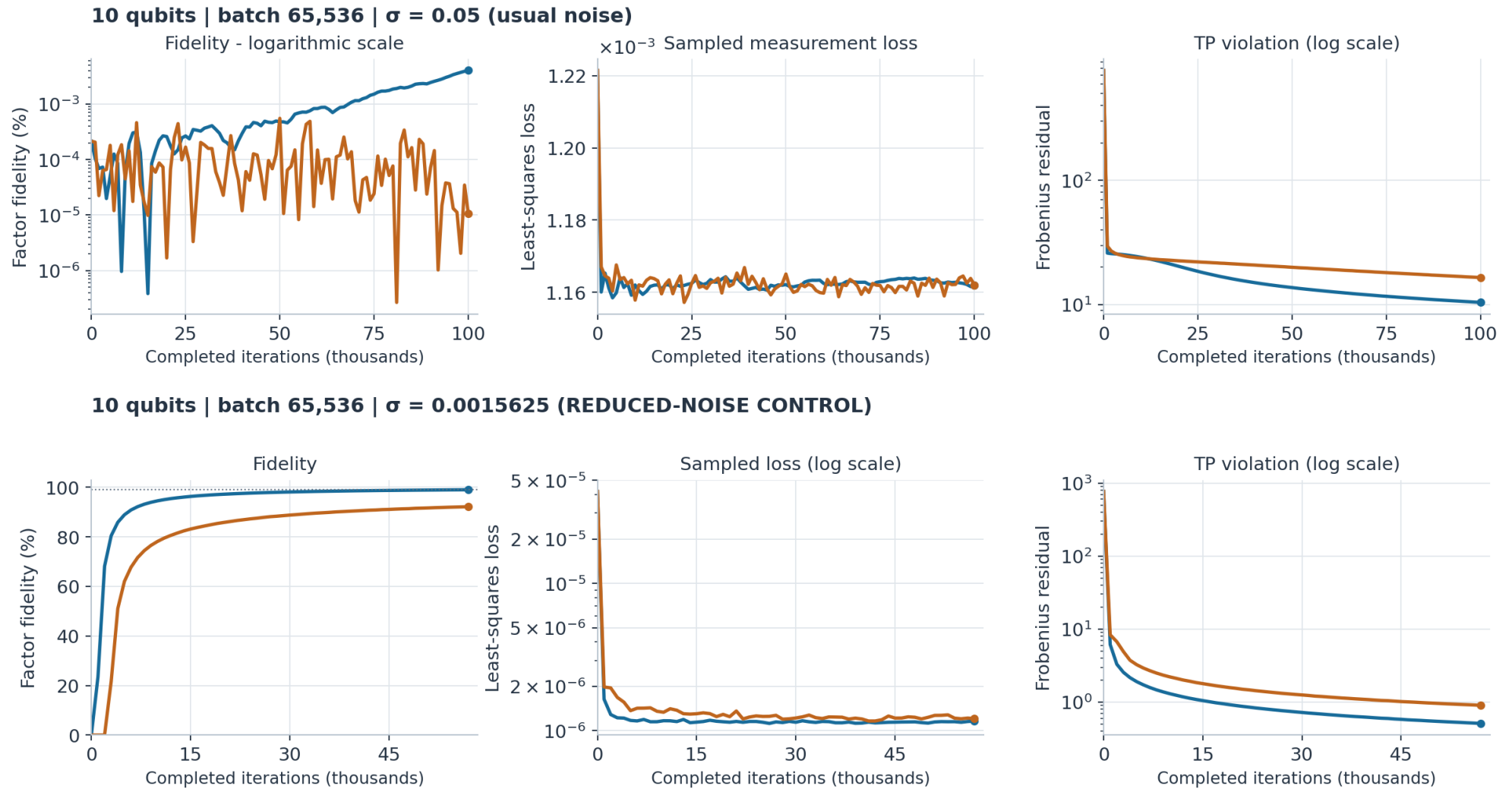


The old B=65,536 trajectory reaches 99.0027%; the new one reaches 90.7749% at the same 148,000 steps. For B=8,192 at 200,000 steps, fidelity falls from 90.8853% to 75.5377%. Both new TP residuals increase.

Fidelity and TP are complete-factor diagnostics. Loss uses 512 fixed sampled rows. Logarithmic loss/TP axes where indicated; dotted fidelity line marks 99% when in range.

Ten qubits: usual noise and reduced-noise control

— Old: step 1, smoothing 1/4, momentum 0.6 — New: step 3/4, smoothing 1/4, momentum 1/2



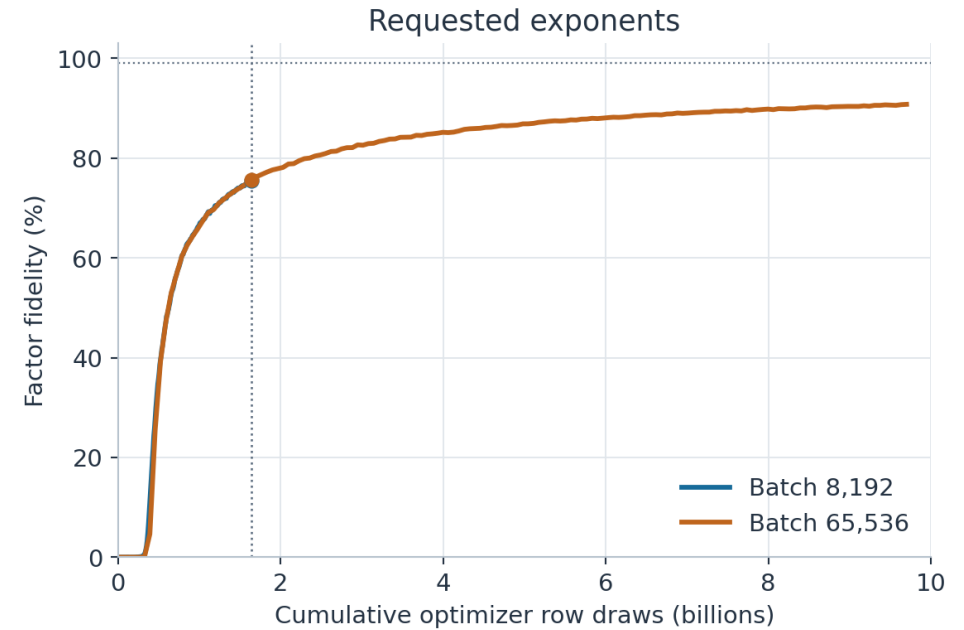
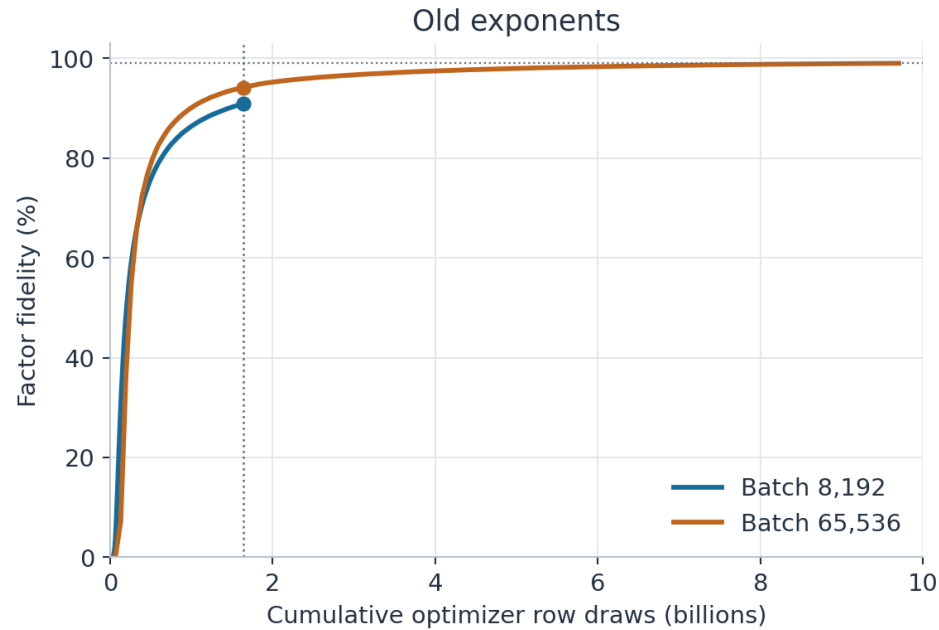
Top: $\sigma=0.05$ gives essentially no recovery under either schedule; the fidelity axis is logarithmic. Bottom: the distinct reduced-noise control, $\sigma=0.0015625$, falls from 99.0045% to 92.1621% at 57,000 steps.

Fidelity and TP are complete-factor diagnostics. Loss uses 512 fixed sampled rows. Logarithmic loss/TP axes where indicated; dotted fidelity line marks 99% when in range.

Eight-qubit comparison at equal row-draw budgets

8 qubits: recovery versus sampled-row budget

$\sigma = 0.05$ | dotted vertical line: 1.6384 billion draws | repeated rows count again



Schedule	Batch	Steps	Fidelity	Sampled loss	TP residual
Old	8,192	200,000	90.8853%	0.001135	0.4835
Old	65,536	25,000	94.1584%	0.001136	0.5153
New	8,192	200,000	75.5377%	0.001145	0.8669
New	65,536	25,000	75.6198%	0.001144	0.8467

All four rows use exactly 1,638,400,000 optimizer row draws (1.48844% of the virtual table size). Under the requested schedules, the two fidelities are nearly equal at this budget. Schedules depend on iteration count, so equal draw budgets do not isolate batch variance or equalize γ , β and ρ .

Best saved checkpoints and reproducibility

New run	Best saved F	Best step	First saved $\geq 99\%$	Optimizer (min)	Runner (min)
n=4, β scale 10	41.9235%	10,000	Not reached	0.027	0.115
n=4, β scale 100	99.6689%	10,000	3,200	0.027	0.115
n=4, β scale 1,000	96.8835%	9,500	Not reached	0.027	0.119
n=4, β scale 10,000	96.3314%	9,500	Not reached	0.027	0.117
n=5, β scale 10	0.9729%	10,000	Not reached	0.037	0.151
n=5, β scale 100	9.1272%	10,000	Not reached	0.036	0.141
n=5, β scale 1,000	93.5486%	9,700	Not reached	0.036	0.143
n=5, β scale 10,000	90.3679%	9,600	Not reached	0.037	0.146
n=8, B=8,192	75.5377%	200,000	Not reached	4.919	10.808
n=8, B=65,536	90.7749%	148,000	Not reached	23.791	62.061
n=10, usual noise	0.000551%	50,000	Not reached	158.217	169.825
n=10, reduced noise*	92.1621%	57,000	Not reached	90.127	96.115

All new runs completed their fixed budgets even after a target crossing. Best values are maxima over saved checkpoints (every 100 updates for n=4/5, every 1,000 for n=8/10). Optimizer timing excludes setup and metrics; runner timing includes preparation, metrics and checkpointing. Neither is whole-job elapsed time.

Validation and execution. All five Slurm job records completed with exit 0. Every case passed independent complete-factor NumPy checks in coefficient and matrix bases, plus dataset/U0/source hash and schedule checks. Final and restart archives are backed up locally; all 132 checksum-listed files match. n=4/5 used V100S 32GB on gpu09; n=8/10 used A100-SXM4 40GB on gpu15, float64/complex128, with GPU autotuning disabled.

Comparison details. Old n=8 traces join exact continuations at 30k (small batch) and 100k (large batch); U0 matches the fresh new run. The old large batch used chunk 10 then 1, while the new run uses previously validated chunk 1 throughout. Old 99% runs stopped at 148k and 57k; new runs retain those budgets without early stopping. Saved diagnostic β can use the next update at an old early-stopped endpoint versus the last update at a fixed endpoint. This one-index difference affects smoothed diagnostics, not F, loss or TP.

Evidence. Campaign: results/qpt/exponents_20260922_01. Comparison tables, all scalar traces and figure scripts are retained there, with command provenance and archive checksums per case. Frozen source: fidelity99_prefetch_validation_20260912_01/source; base revision ba85bff73c30. Jobs: 1925109 and 1925110_0..3. No algorithms/defaults, Git revisions or existing reports were changed for this comparison. The prior coauthor PDF is preserved.